University of Pretoria - Department of Mathematics and Applied Mathematics
Universiteit van Pretoria - Departement Wiskunde en Toegepaste Wiskunde

CALCULUS WTW158
SEMESTER TEST 1 / SEMESTERTOETS 1

Date / Datum: March / Maart 2014 Marks / Punte : 48 Time / Tyd : $2\frac{1}{2}$ hours / uur

MARKS FOR SECTION B OUT OF 25 **PUNTE VIR AFDELING B UIT 25**	8 ✓

SURNAME / VAN

FIRST NAMES / VOORNAME ____________________

STUDENT NUMBER
STUDENTENOMMER

SIGNATURE
HANDTEKENING ____________________

CIRCLE YOUR GROUP NUMBER: ENGLISH GROUPS

4	5	6	7	8	9
Dr Mutangadura	Ms Dinga	Dr van der Hoff	Dr Chin	Dr Moubandjo	Dr Chin
Mo 7:30 Eng III-3	Mo 7:30 Centenary 5	Mo 7:30 Thuto 3-1	Mo 7:30 VdBijl	Mo 8:30 Eng III-4	Mo 12:30 Eng III-6

OMKRING JOU LESINGGROEPNOMMER AFRIKAANSE GROEPE

1	2	3
Me Verwey	Me Dreyer	Me Dreyer
Ma 7:30 Ing III-1	Ma 13:30 Ing I 1-1	Ma 7:30 Ing III-2

Internal Examiners / Interne Eksaminatore	**WTW 158 Lecturers**
External Examiner / Eksterne Eksaminator	**Ms / Me R Moller**

READ THE FOLLOWING INSTRUCTIONS

1. The paper consists of pages 1 to 14. Check whether your paper is complete.
2. **No calculator may be used.**
3. No question paper may be removed from the venue.
4. Any queries about the marking must be done within three days after the tests have been handed back.

LEES DIE VOLGENDE INSTRUKSIES

1. Die vraestel bestaan uit bladsye 1 tot 14. Kontroleer of u vraestel volledig is.
2. **Geen sakrekenaar mag gebruik word nie.**
3. Geen vraestel mag uit die lokaal geneem word nie.
4. Enige navrae oor die nasienwerk moet binne drie dae nadat die toetse teruggegee is, gedoen word.

SECTION A: MULTIPLE CHOICE QUESTIONS - 23 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. YOU HAVE TO START WITH SECTION A, AS YOU HAVE TO HAND IN THE OPTIC READER FORM AFTER 75 MINUTES.
2. As you may not erase wrong answers on the optic reader form, first circle your answers on this paper and only mark the answers on the form when you are certain of your choice.
3. Do all scribbling on the facing page. It will not be marked.
4. Use a soft pencil to write on the optic reader form.
5. Use **SIDE 2** of the optic reader form.
   Fill in your personal information in pencil on side 2.
   Fill in your student number from top to bottom and then code it.
6. **You may start with section B as soon as you have marked your answers on the optic reader form.**

AFDELING A : MEERVOUDIGE KEUSEVRAE - 23 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. U MOET MET AFDELING A BEGIN WANT U MOET DIE MERKLEESVORM NA 75 MINUTE INHANDIG.
2. U mag nie verkeerde antwoorde uitvee op die merkleesvorm nie. Omkring dus u antwoorde eers in hierdie vraestel en merk die antwoorde op die vorm as u seker is van u keuse.
3. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
4. Gebruik 'n sagte potlood om op die merkleesvorm te skryf.
5. Gebruik **KANT 2** van die merkleesvorm.
   Vul u persoonlike inligting in potlood in op kant 2.
   Vul u studentenommer van bo na onder in en kodeer dit.
6. **U kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.**

Question 1 / Vraag 1

Consider the function / Beskou die funksie

$$F(x) = \frac{x+4}{\sqrt{x^2+3x-4}}$$

The domain of F is given by / Die definisieversameling van F word gegee deur

(a)	$(-4,1)$	(b)	$\mathbb{R}$	(c)	$(-\infty,-4]\cup[1,\infty)$	(d)	$[-4,1]$
(e)	$(-\infty,-4)\cup(1,\infty)$	(f)	$(-\infty,-1)\cup(4,\infty)$	(g)	None of these / Geen van hierdie		

Question 2 / Vraag 2

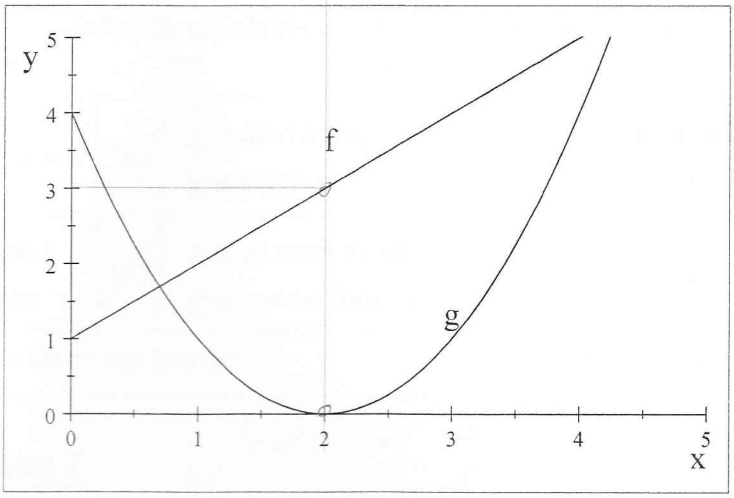
The solution set of $(4-x)|2-3x|>0$ is

Die oplossingsversameling van $(4-x)|2-3x|>0$ is

(a)	ϕ	(b)	$(-\infty,\frac{2}{3})\cup(\frac{2}{3},4)$	(c)	$\mathbb{R}$	(d)	$(-\infty,\frac{2}{3})\cup(4,\infty)$
(e)	$(-\infty,\frac{2}{3})\cap(4,\infty)$	(f)	$(\frac{2}{3},4)$	(g)	None of these / Geen van hierdie		

Question 3 / Vraag 3

Consider the graphs of functions f and g given below. / Beskou die grafieke van die funksies f en g gegee hieronder.



$(f \circ g)(2) =$ $f(g(x))$

(a)	0	(b)	1	(c)	2	(d)	3
(e)	4	(f)	5	(g)	None of these / Geen van hierdie		

Question 4 / Vraag 4

Consider / Beskou $F(x) = \sec^4(x-4)$.
If $F(x) = (f \circ g \circ h)(x)$, then the functions f, g and h can be represented as follows:
Indien $F(x) = (f \circ g \circ h)(x)$ dan kan die funksies f, g en h as volg voorgestel word:

(a)	$f(x) = x - 4$ $g(x) = \sec x$ $h(x) = x^4$ $\sec(x-4)^4$	(b)	$f(x) = x - 4$ $g(x) = x^4$ $h(x) = \sec x$	(c)	$f(x) = x^4$ $g(x) = \sec x$ $h(x) = x - 4$	(d)	$f(x) = x^4$ $g(x) = x - 4$ $h(x) = \sec x$
(e)	$f(x) = \sec x$ $g(x) = x - 4$ $h(x) = x^4$	(f)	$f(x) = \sec x$ $g(x) = x^4$ $h(x) = x - 4$	(g)	None of these / Geen van hierdie		

Question 5 / Vraag 5

Which of the given functions are odd functions?
Watter van die volgende funksies is onewe funksies?
$y_1 = \frac{x-1}{x}$, $y_2 = x^3 + 2x$, $y_3 = x^3 + 2$

(a)	y_1 and/en y_2	(b)	y_1 and/en y_3	(c)	Only/Slegs y_1	(d)	Only/Slegs y_2
(e)	y_2 and/en y_3	(f)	Only/Slegs y_3	(g)	None of them / Geeneen van hulle		

Question 6 / Vraag 6

Let g be a function defined for all real numbers. In which of the following cases does g not have an inverse function.

Laat g 'n funksie wees gedefinieer vir alle reële getalle. In watter van die volgende gevalle het g nie 'n inverse funksie nie?

(a)	$g(x) = ax + b, a \neq 0$	(b)	g is increasing / g is stygend	(c)	g is decreasing / g is dalend
(d)	g is one-to-one / g is een-tot-een	(e)	g is an even function / g is 'n ewe funksie		
(f)	None of these / Geen van hierdie				

Question 7 / Vraag 7

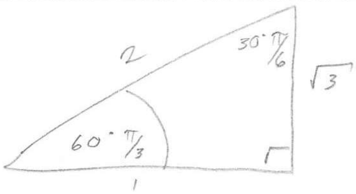
sin(sin⁻¹ x)

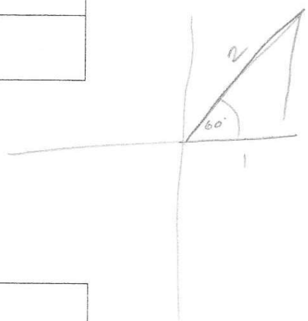
Let $y = f(x) = \sin(\arcsin x)$. The domain of f is

Laat $y = f(x) = \sin(\text{bgsin}\, x)$. Die definisieversameling van f is

(a)	~~$\{x \mid 0 < x \leq 1\}$~~	(b)	$\{x \mid -1 \leq x \leq 1\}$	(c)	~~$\{x \mid 0 \leq x \leq 1\}$~~
(d)	$\{x \mid -\frac{\pi}{2} < x < \frac{\pi}{2}\}$	(e)	$\{x \mid -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}\}$		
(f)	~~$\{x \mid 0 < x < \pi\}$~~	(g)	None of these / Geen van hierdie		

Question 8 / Vraag 8





$\cos(\frac{7\pi}{3}) =$

(a)	-1	(b)	$\frac{\sqrt{3}}{2}$	(c)	$-\frac{\sqrt{3}}{2}$	(d)	$-\frac{1}{2}$
(e)	1	(f)	$\frac{1}{2}$	(g)	None of these / Geen van hierdie		

Question 9 / Vraag 9

If $\tan x = \frac{1}{3}, \pi < x < \frac{3\pi}{2}$, then $\cos x =$

As $\tan x = \frac{1}{3}, \pi < x < \frac{3\pi}{2}$, then $\cos x =$

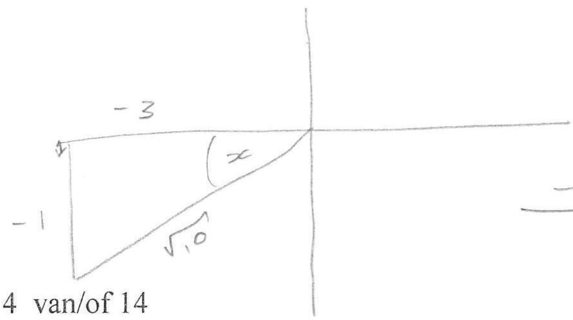
(a)	3	(b)	$\frac{1}{\sqrt{10}}$	(c)	$\frac{-1}{\sqrt{10}}$	(d)	$\frac{3}{\sqrt{10}}$
(e)	$\frac{-3}{\sqrt{10}}$	(f)	$\frac{3}{\sqrt{8}}$	(g)	None of these / Geen van hierdie		

$180 < x < 270$

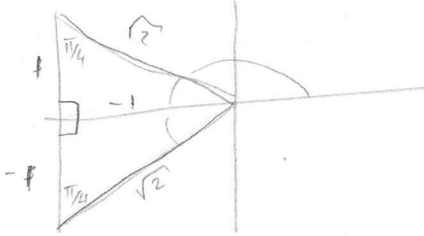
$\sqrt{(-1)^2 + (-3)^2}$

$\sqrt{1+9}$

$\sqrt{10}$



$\frac{-3}{\sqrt{10}}$



Question 10 / Vraag 10

$\arccos\left(-\frac{1}{\sqrt{2}}\right) =$ / $\text{bgcos}\left(-\frac{1}{\sqrt{2}}\right) =$

$\cancel{(a)}$	$\cancel{\frac{\pi}{2}}$	(b)	$-\frac{\pi}{4}$	(c)	$\frac{3\pi}{4}$
$\cancel{(d)}$	$\cancel{-\frac{\pi}{2}}$	(e)	$-\frac{3\pi}{4}$		
$\cancel{(f)}$	$\cancel{0}$	~~(g)~~	~~None of these / Geen van hierdie~~		

Question 11 / Vraag 11

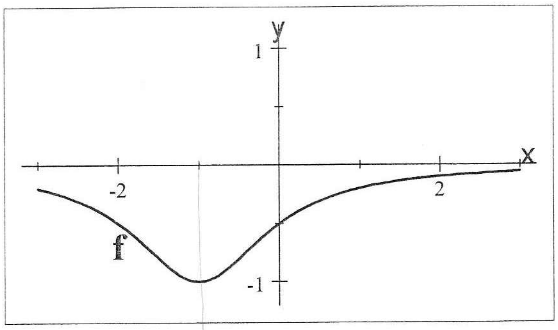
If $f(x) = 2e^{-x}$ then $f^{-1}(x) =$

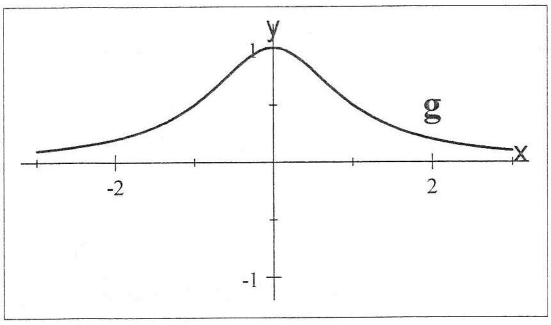
As $f(x) = 2e^{-x}$ dan is $f^{-1}(x) =$

(a)	$\ln\left(\frac{x}{2}\right)$	(b)	$\ln\left(\frac{2}{x}\right)$	(c)	$\frac{1}{2}\ln x$	(d)	$\sqrt{\ln x}$
(e)	$\ln(2-x)$	(f)	$\ln(x-2)$	(g)	None of these / Geen van hierdie		

Question 12 / Vraag 12

Consider the graphs of f and g given below. / Beskou die grafieke van f en g gegee hieronder.





$g(x) =$

(a)	$-f(x-1)$	(b)	$f(x+1)$	(c)	$-f(x+1)$	(d)	$-f(x)-1$
(e)	$-f(x)+1$	(f)	$f(x)-1$	(g)	None of these / Geen van hierdie		

$f(x)$

$-f(x-1)$

Question 13 / Vraag 13

$\sin\left(\frac{3\pi}{2}\right) + \tan\left(\frac{7\pi}{4}\right) =$

(a)	$\frac{1}{2}$	(b)	$-\frac{1}{2}$	(c)	-1	(d)	1
(e)	-2	(f)	2	(g)	None of these / Geen van hierdie		

$5e^{x+\ln 2} = 2$
$e^{x+\ln 2} = 2/5$
$x + \ln 2 = \ln 2/5$
$x = \ln 2/5 - \ln 2$
$x = \ln$

$\frac{\left(\frac{2}{5}\right)}{\left(\frac{2}{1}\right)} = \left(\frac{2}{5}\right)\left(\frac{1}{2}\right) = \frac{4}{5}$

Question 14 / Vraag 14

$(\ln e)^3 - \ln(e^3) + \ln \frac{1}{e} =$

(a)	-1	(b)	1	(c)	-2	(d)	2
(e)	-3	(f)	3	(g)	None of these / Geen van hierdie		

$8e^{-x} = \frac{1}{e}$
$e^{-x} = \frac{8}{e}$
$-x = \ln \frac{8}{e}$
$x = -\ln \frac{8}{e}$

Question 15 / Vraag 15

If $5e^{x+\ln 2} = 2$, then $x =$ / As $5e^{x+\ln 2} = 2$, dan is $x =$

(a)	$-\ln 2$	(b)	$\ln 2$	(c)	$-\ln 5$	(d)	$\ln 5$
(e)	$-\ln \frac{2}{5}$	(f)	$\ln \frac{2}{5}$	(g)	None of these / Geen van hierdie		

$8e^{-x} = \frac{1}{e}$
$e^{-x} = (8e)^{-1}$
$-x = \ln(8e)^{-1}$
$-\ln 8e$
$x = \ln 8e$

Question 16 / Vraag 16

4

If $8e^{-x} = \frac{1}{e}$ then $x =$ / As $8e^{-x} = \frac{1}{e}$, dan is $x =$

(a)	$\ln 8 - 1$	(b)	$3\ln 2 + 1$	(c)	$\ln 4 - 1$	(d)	$2\ln 2 + 1$
(e)	1	(f)	0	(g)	None of these / Geen van hierdie		

e^{-8x}

Question 17 / Vraag 17

$\lim_{x \to 3} \frac{x-3}{x^2 - 2x - 3} =$

$\cancel{(x-3)}(x+1)$

$\frac{1}{x+1} = \frac{1}{4}$

(a)	0	(b)	1	(c)	$\frac{1}{4}$	(d)	$\frac{1}{2}$
(e)	4	(f)	∞	(g)	None of these / Geen van hierdie		

$(0)\ \frac{1}{\cos(0)}$

$x\ \frac{1}{\cos}(x)$

Question 18 / Vraag 18

$\lim_{x \to 0} x\operatorname{cosec}(x) =$

(a)	$-\infty$	(b)	-1	(c)	0	(d)	1
(e)	∞	(f)	Does not exist / Bestaan nie	(g)	None of these / Geen van hierdie		

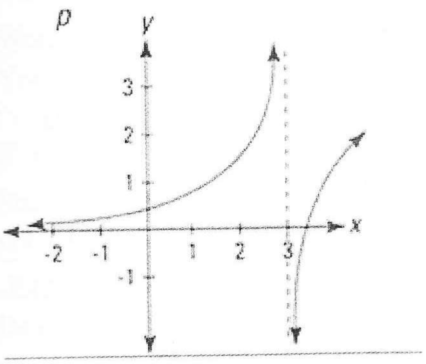
Question 19 / Vraag 19

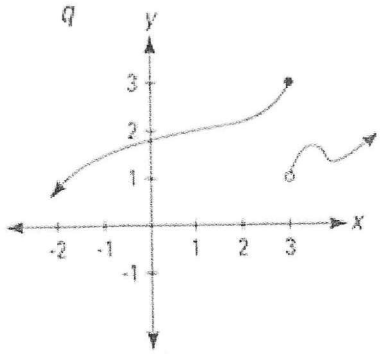
If / As $\ln(x^2) - 4 = 0$ then / dan is $x =$

(a)	$\pm e^2$	(b)	$-e^2$	(c)	e^2	(d)	1
(e)	-1	(f)	± 1	(g)	None of these / Geen van hierdie		

Die onderstaande inligting is van toepassing op Vrae 20 en 21.
The information below applies to Questions 20 and 21.

The graphs of the functions p and q are given below. / Die grafieke van die funksies p en q is gegee hieronder.





Question 20 / Vraag 20

$\lim_{x \to 3^-} p(x) =$

(a)	3	(b)	$-\infty$	(c)	$+\infty$	(d)	1
(e)	0	(f)	$\pm\infty$	(g)	None of these / Geen van hierdie		

Question 21 / Vraag 21

$\lim_{x \to 3^+} q(x) =$

(a)	3	(b)	$-\infty$	(c)	∞	(d)	1
(e)	1.5	(f)	Does not exist / Bestaan nie	(g)	None of these / Geen van hierdie		

Question 22 / Vraag 22

$$\lim_{x \to 2^+} \frac{x-1}{x-2} =$$

$\frac{2,5}{} \quad \frac{1,5}{0,5} = 3 \qquad \frac{2,1}{} \quad \frac{1,1}{0,1} = 11$

(a)	0	(b)	∞	(c)	5	(d)	$-\infty$
(e)	4	(f)	$\frac{1}{2}$	(g)	None of these / Geen van hierdie		

Question 23 / Vraag 23

$$\lim_{x \to 2^-} \frac{x}{x^2(x-2)} =$$

(a)	0	(b)	∞	(c)	5	(d)	$-\infty$
(e)	4	(f)	1	(g)	None of these / Geen van hierdie		

23

SECTION B: 25 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer all the following questions on this paper **in ink**.
2. Work done in pencil or in red ink will not be marked.
3. You are not allowed to use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked. If you need more space for an answer, use the facing page and indicate it clearly **by framing it**.
5. Show all steps and calculations and explain your answers.

AFDELING B: 25 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel **in ink**.
2. Werk wat in potlood of rooi ink beantwoord is, word nie nagesien nie.
3. U mag nie korrigeer-vloeistof ("Tipp-Ex") gebruik nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie. As u nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan **deur 'n raam daarom te trek**.
5. Toon alle stappe en berekeninge en verduidelik u antwoorde.

Question 1 / Vraag 1

Solve for x if $\frac{1}{3-x} \le 2$.

Los op vir x as $\frac{1}{3-x} \le 2$.

Lcd!

$\frac{1}{3-x} \le 2$

$1 \le (2)(3-x)$

$1 \le 6 - 2x$

$1 - 6 \le -2x$

$-5 \le -2x$

$5 \ge 2x$

$\frac{5}{2} \ge x$

$\therefore x \le \frac{5}{2}$

$x \; (-\infty ; \frac{5}{2}]$

0

3

Question 2 / Vraag 2

(i) Complete the following: / Voltooi die volgende:

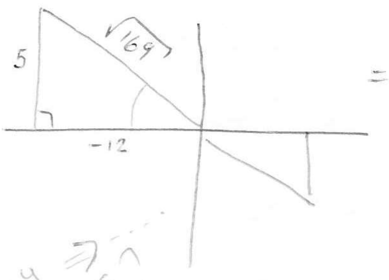
$y = \tan^{-1} x \Leftrightarrow$ $x = \tan y$ for all / vir alle $x \in$ $(-\frac{\pi}{2} ; \frac{\pi}{2})$

and for all / en vir alle $y \in$ $\mathbb{R}$

1

2

(ii) Evaluate $\sin\left(\tan^{-1}\left(-\frac{5}{12}\right)\right)$.
Bepaal $\sin\left(\tan^{-1}\left(-\frac{5}{12}\right)\right)$.



$\sqrt{(5)^2 + (-12)^2}$

$= \sqrt{25 + 144}$

$= \sqrt{169}$

$y \Rightarrow y \in ?$

$\sin\left(\tan^{-1}\left(-\frac{5}{12}\right)\right) = \frac{-5}{\sqrt{169}}$

$\frac{1}{2}$

2

Question 3 / Vraag 3

(i) For which values of x is $\ln(2x+1) + \ln x > 0$?

Vir watter waardes van x is $\ln(2x+1) + \ln x > 0$?

$\ln(2x+1) + \ln x > 0$

$\ln[(2x+1)(x)] > 0$

$\ln(2x^2+x) > 0$

$2x^2 + x > e^0$

$2x^2 + x > 1$

$2x^2 + x - 1 > 0$

$(2x-1)(x+1) > 0$

$2x - 1 = 0$ $\quad$ $x + 1 = 0$

$x = 1/2$ $\quad$ $x = -1$

$(-\infty; -1) \cup (1/2; +\infty)$?? How

$1\frac{1}{2}$

4

Question 4 / Vraag 4

(i) Use the definition of the absolute value to complete the following. /

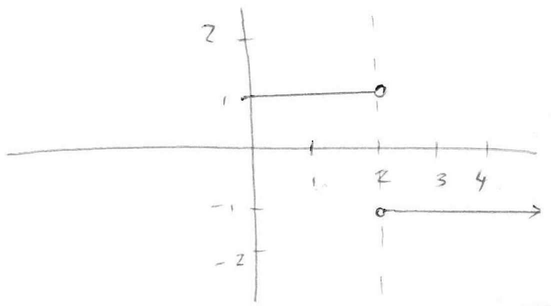
Gebruik die definisie van die absolute waarde en voltooi die volgende.

$$|2-x| = \begin{cases} 2-x & \text{if / as } \quad 2-x \geq 0 \\ -(2-x) & \text{if / as } \quad 2-x < 0 \end{cases}$$

$x \geq ?$
$x < ?$

(ii) Use 4(i) to determine / Gebruik 4(i) en bepaal

$\lim\limits_{x \to 2} \dfrac{2-x}{|2-x|}$ (if it exists / indien dit bestaan)



$$|2-x| = \begin{cases} 2-x & \text{if } 2-x \geq 0 \\ -2-x & \text{if } 2-x < 0 \end{cases}$$

$2-(2) = 0 \quad \therefore \quad |2-x| = 2-x$

$$\lim_{x \to 2} \frac{2-x}{|2-x|} = \lim_{x \to 2} \frac{2-x}{2-x} = \frac{2-(2)}{2-(2)} = \frac{0}{0}$$

THE LIMIT DOES NOT EXIST

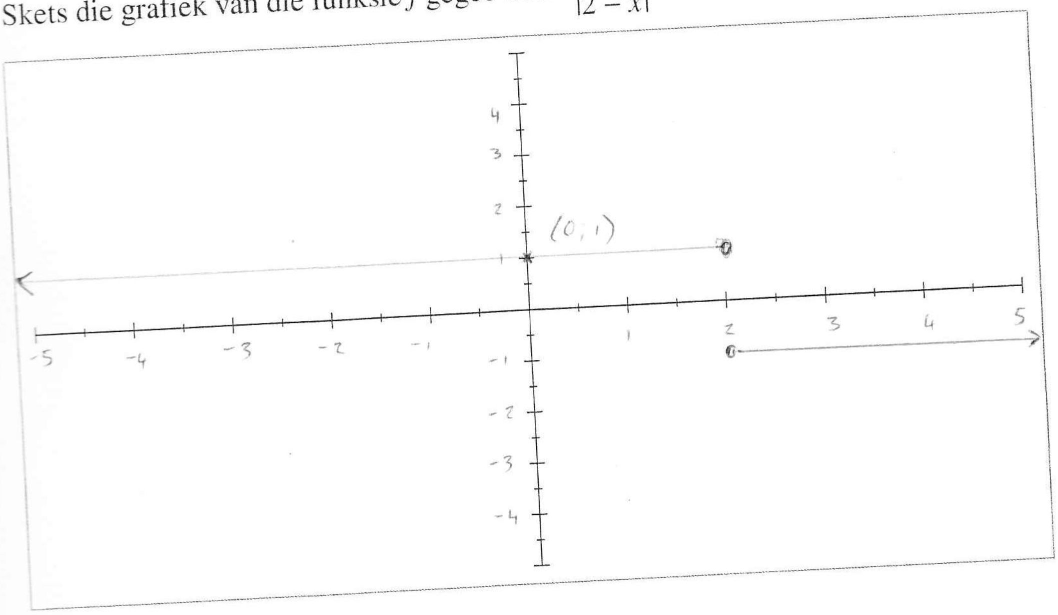
$\lim\limits_{x \to 2^-} \dfrac{2-x}{|2-x|} =$

$\lim\limits_{x \to 2^+} \dfrac{2-x}{|2-x|} =$

$2^- \neq 2^+$

(iii) Sketch the graph of the function f given by $f(x) = \dfrac{2-x}{|2-x|}$. Indicate the intercepts with the axes.

Skets die grafiek van die funksie f gegee deur $\dfrac{2-x}{|2-x|}$. Dui die snypunte met die asse aan.



Question 5 / Vraag 5

Given / Gegee $f(x) = \frac{1}{x}$ and/en $g(x) = \frac{x}{1+x}$.

Determine $(f \circ g)(x)$ and state the domain of $f \circ g$.

Bepaal $(f \circ g)(x)$ en gee die definisieversameling van $f \circ g$.

$$(f \circ g)(x) = f(g(x))$$

$$= \frac{1}{\left(\frac{x}{1+x}\right)}$$

$$= \left(\frac{1}{1}\right) \times \left(\frac{1+x}{x}\right)$$

$$= \left(\frac{1+x}{x}\right)$$

DOMAIN $f \circ g$ IS WHERE BOTH f AND g IS DEFINED.

DOMAIN f: $x \neq 0$ $\quad (-\infty; 0) \cup (0; \infty)$

DOMAIN g: $1+x \neq 0$ $\quad (-\infty; -1) \cup (-1; \infty)$

$x \neq -1$

$\therefore$ DOMAIN $f \circ g$: ~~$(-\infty)$~~

$(-\infty; -1) \cup (-1; 0) \cup (0; \infty)$

$x \neq 0$ AND $x \neq -1$

3

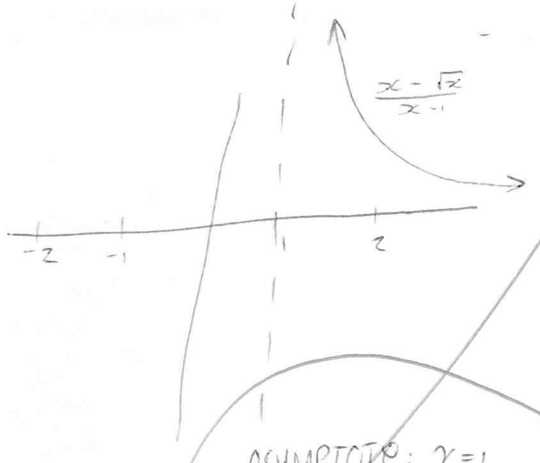
3

Question 6 / Vraag 6

Determine / Bepaal

(i) $\lim_{x \to 1} \frac{x - \sqrt{x}}{x - 1}$

$\frac{(1) - \sqrt{1}}{1 - 1} = \frac{0}{0}$



$x - 1 \neq 0$

$x \neq 1$

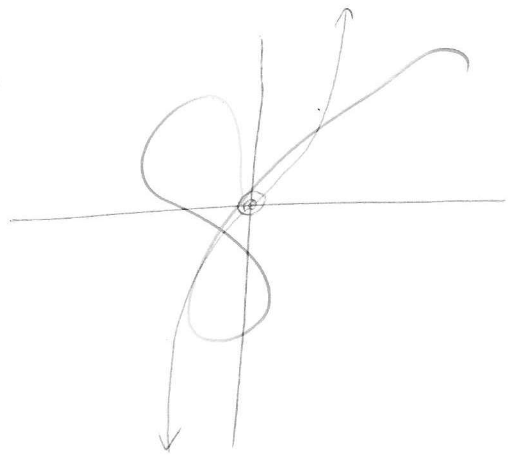
ASYMPTOTE: $x=1$

$\lim_{x \to 1^+} \frac{x - \sqrt{x}}{x - 1} = +\infty$

3

(ii) $\lim_{x \to 0} \frac{\tan 4x}{x}$

$\tan(0) = 0$



$\lim_{x \to 0} \frac{\tan 4x}{x} = 0$

1

Question 7 / Vraag 7

Given / Gegee $f(x) = \dfrac{2-x}{x^2-3x+2}$

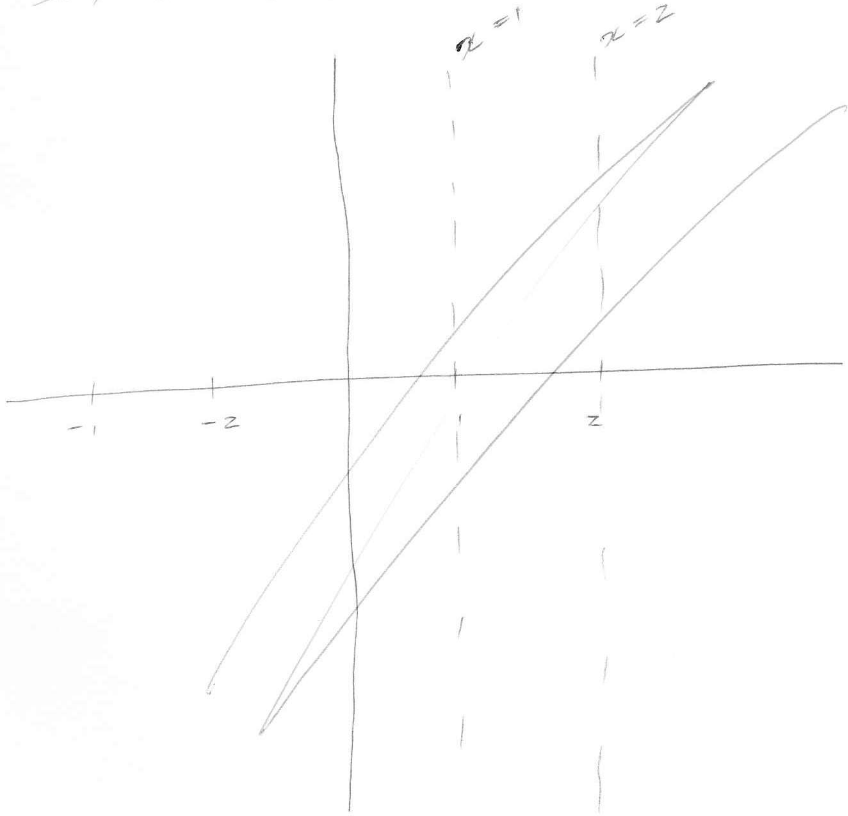
Determine the vertical asymptotes of f (if they exist).

Bepaal die vertikale asimptote van f (indien hulle bestaan).

$x^2 - 3x + 2 \neq 0$

$(x-2)(x-1) \neq 0$

$x \neq 2 \quad x \neq 1$



THE VERTICAL ASYMPTOTES ARE

~~x ≠ 1~~ AND $\cancel{x = 2}$

$x = 1$

3

0